



15th RDKit UGM Agenda v1.6

16-18 September 2026

Darmstadt, Germany

The UGM will be an in-person event with the presentations lived-streamed via zoom. If you have registered for the virtual version of the UGM, you will get an email a day or two before it starts with the zoom link.

cd

In past years we have used discord for online discussion about the UGM. After having gotten feedback that discord is being blocked by some companies, this time we're trying out Zulip:

<https://tinyurl.com/yvk7nncc>

The Github repo for this year is here: https://github.com/rdkit/UGM_2026

The most recent version of this document, logistics information, and other practical details can be found in the [github repo](#).

Wednesday 16 September

Time (CEST)	
08:30 - 09:00	Check-in / registration
09:00 - 10:00	Greg Landrum Welcome, State of the RDKit, and What's New part 1
10:00 - 10:30	Niels Maeder ETKDGv4: Next generation conformer generator in the RDKit
10:30 - 11:30	Coffee and posters
11:30 - 11:50	Paolo Tosco Pep-tidying: challenges in curating heterogeneous peptide datasets
11:50 - 12:10	Jakub Adamczyk MolPILE - large-scale dataset for molecular representation learning
12:10 - 12:30	Kevin Boyd Substructure Search on the GPU with nvMolKit
12:30 - 13:30	Lunch and posters
13:30 - 14:00	Lightning talks 1
14:00 - 14:30	David Cosgrove Shape-based similarity searching in synthon space.
14:30 - 15:00	Alister Goodfellow Embedding and stereochemistry for transition metal complexes
15:00 - 16:00	Coffee and posters
16:00 - 16:30	Hannah Baumann Making alchemical free energy calculations practical at scale with OpenFE
16:30 - 17:00	Maciej Wójcikowski Accelerating massive substructure matching with roaring bitmaps
17:00 - 17:30	Masaya Fujii Dynamic Attention-Based Substructure Hierarchy for surface area assignment
19:00 - 23:00	Conference Dinner at "Das Roeders" - https://maps.app.goo.gl/jDEXjroMJspm7iWA6

Thursday 17 September

Time (CEST)	
08:30 - 09:00	Check-in / registration
09:00 - 09:30	What's new 2
09:30 - 10:00	Andrew Dalke Approximating Count Fingerprints with Binary
10:00 - 10:30	Michel Lim Understanding the role of aromaticity in tautomeric equilibria
10:30 - 11:30	Coffee and posters
11:30 - 11:50	Hagen Muenkler qm_atlas: End-to-End Automation of Quantum Chemistry Workflows for Molecular Property Prediction
11:50 - 12:10	Piotr Ludynia Particularly strong and robust performance of fingerprint-based approach for peptide function prediction
12:10 - 12:30	Thomas Wolf Mother ML: A Modular Framework for Reproducible Cheminformatics
12:30 - 13:30	Lunch and posters
13:30 - 14:00	Lightning Talks 2
14:00 - 14:30	Gregori Gerebtzoff Comprehensive bio-transformation analysis of oligonucleotides with RDKit
14:30 - 15:00	Martin Fitzner Chemical Encodings for Bayesian Optimization in the Real World
15:00 - 15:30	Coffee and posters
15:30 - 16:00	What's new 3
16:00 - 16:30	Michael Backenköhler ChemICL: Uncertainty-Aware Bioactivity Modelling at Scale
16:30 - 17:00	Roger Sayle Roger's talk
17:00 - 17:30	Wrap up
19:00 - 23:00	Dinner: small groups at local restaurants

Lightning Talks

Session 1

Oscar Palomino-Hernandez

Anna Santura

Carl Schiebroek

Thomas Evangelidis

Hideyoshi Fuji

Rostislav Fedorov

Session 2

Esben Jannik Bjerrum

Wim Dehaen

Jakub Adamczyk

Jonathan Zheng

Adam Staniszewski

Rachel Walker

Friday 18 September

Time (CEST)	
9:00 - 9:30	Check-in / registration
9:30 - 16:00	Hackathon! and Workshops

The hackathon is, as always, spontaneously organized and informal.

In addition to the normal hackathon, we will have three workshops/tutorials on Friday morning:

1. Hands-On KNIME: New Features, Cheminformatics Workflows & Solution Building

Join us for an interactive KNIME session designed to expand your workflow building skills and solve real-world problems. We'll start with a quick demo highlighting the latest browser-based workflow editor, new nodes, and UI updates. Then, the floor is yours: bring your own use cases for live problem solving, integrating your Python script into a workflow, ask any technical or extension-development questions, or practice with pre-made chemistry data exercises covering either data import, MCS finding, R-group decomposition, and fingerprint similarity clustering.

2. CReM: Flexible and Customizable Molecular Design through Chemically Reasonable Mutations

The tutorial will introduce the fundamental principles of CReM, including its recent extension for ring-system generation and key features with practical implications, such as control over synthetic accessibility. Participants will be guided through the complete workflow, from fragment database preparation to structure generation. We will also briefly overview CReM-based approaches for 3D pharmacophore- and docking-guided molecular design followed up with use cases.

3. Bayesian Statistical Testing for QSAR Modelling

When you benchmark QSAR models, does your new architecture really beat the ECFP baseline, or is the gap just the noise of your evaluation? Standard significance tests do not answer that directly, whereas Bayesian tests estimate the probability that one model is better, that the models are practically equivalent, or that the data is not yet decisive. This workshop explains how Bayesian testing differs from null hypothesis significance testing, introduces tests such as the hierarchical correlated t-test and the Bayesian Bradley-Terry model, and shows how to run them with a few lines of Python.

Sponsors

The RDKit UGM can only take place in its current form thanks to the generous financial support of our sponsors.

Organizing sponsor



Gold



Schrödinger



Silver



Open for Innovation

KNIME



datagrok

Bronze



CDD.VAULT®
Complexity Simplified